Module 2 Measurable Learning Objectives:

A. Students will draw and label (model) the infrastructure for a simple drinking water distribution system from surface water source to household.

B. Students will draw and label (model) the infrastructure for a simple drinking water distribution system from groundwater source to household.

C. Students will create a table which compares and contrasts the two main sources of drinking water, including their treatment (i.e., surface water and groundwater).

D. Students will draw and label (model) the infrastructure for a simple decentralized single household septic system from clean water tap (e.g., faucet, toilet tank, shower head) to leach field effluent.

E. Students will draw and label (model) the infrastructure for a simple centralized sanitary sewer system from clean water tap (e.g., faucet, toilet tank, shower head) to treatment plant effluent.

F. Students will create a table which compares and contrasts centralized and decentralized wastewater systems, including reasons why each is used.

G. Based on what they can learn locally, students will draw and label (model) a simple map of their local drinking water system and the wastewater infrastructure used by their own household – from drinking water source to effluent, and be prepared to answer questions about anything on their map.

H. Given different items which can be poured down a drain (e.g., dishwater, vinegar, cooking oil/grease, bleach) or flushed down a toilet (e.g., human waste (urine/feces), toilet paper, baby wipes, stuffed animals, Stretch Armstrong), students will analyze each item and predict what is likely to happen to the item in the wastewater system and what effect, if any, the item may have on the system.

I. Given a problem scenario from the module, students will solve the problem and report on their solution by completing the following sub-objectives:

    a. Students will clarify the problem given

    b. Students will predict potential outcomes if the problem is not addressed

    c. Students will predict the source or sources of the problem

d. After selecting one problem source to explore, students will gather multiple (more than 5) resources to research the source of the problem (suggest providing students resources but also allowing students to use internet for research)

e. Students will evaluate the resources gathered to select *at least three* credible resources to use in their analysis

f. Students will analyze (compare/contrast) the selected information to produce at least three alternate solutions

g. Students will evaluate the alternate solutions to determine the solution that solves the problem 1) immediately, and 2) for the long term

h. Students will predict appropriate means to prevent the problem from occurring again

i. Students will report on their investigative process and findings (see Summative/ Deliverables)